RUMAISA AZEEM

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Summary

Computer Science student with over two years of professional software engineering experience, strong C++/Python/JavaScript skills, and published research in AI ethics. Proven leader and innovator with award-winning projects, class-topping academic record, and a passion for building scalable, efficient and innovative software.

Education

KING'S COLLEGE LONDON **2022 – PRESENT**

BSc Computer Science with Management & a Year in Industry

- Grade: 1st Class Honours
- Twice awarded as student with top results in degree cohort (Computer Science with Management Prize 22-23, 23-24)
- Relevant modules: Database Systems, Data Structures & Algorithms, AI & Machine Learning, Functional Programming, Operating Systems & Concurrency, Mathematical Foundations of Computing
- Elected Student Representative for a cohort of 300+ students, responsible for delivering actionable feedback to the Informatics Department and collaborating with them to improve student experience

MANCHESTER HIGH SCHOOL FOR GIRLS **2015 – 2022**

*A-Levels – A*A*A*A**

- A-level subjects: Computer Science, Mathematics, Further Mathematics, Physics
- Awards: School Computer Science Award (2022, 2020), Christine Lee Jones Prize for Leadership & Initiative (2022), Merlyn Lowther Scholarship Award (2020), Academic Scholarship (2015-2022)

Experience

SOFTWARE ENGINEER INTERN (YEAR IN INDUSTRY) **JUL 2024 – JUL 2025**

Hewlett Packard Enterprise

- Independently investigated backup application start time – developed a lightweight simulation of the app's deduplication algorithm reducing testing time by 97%, presented findings to seniors to inform further development and documented actionable insights for external teams to resolve active customer support cases (C++, Python).
- Developed features for and maintained an internal performance-testing web application used by 30+ developers, reducing hours of manual effort required by other engineers and improving accuracy of pre-release testing (Python Flask, TypeScript with React/Redux).
- Architected and implemented cloud storage command line support tools for AWS, GCP, and Azure, shipped on HPE's latest backup appliance (C++).
- Created a novel data profiling method with security and customer profiling applications, winning HPE's Bristol Hack Days and subsequently co-authoring intellectual property disclosures to be submitted as patents.
- Led the team's transition to a Docker-based development environment, delivering automated setup scripts and onboarding documentation.
- Maintained GitHub Actions CICD pipelines to automate building and testing workflows, removing hours of weekly manual testing.
- Recognised as 'Intern of the Quarter for Business Impact' and retained as a part-time intern after the placement year for standout technical contributions.

RESEARCH ASSISTANT **JUN 2023 – DEC 2024**

Responsible Robotics & AI Lab, King's College London

- Co-authored research paper: 'LLM-Driven Robots Risk Enacting Discrimination, Violence, and Unlawful Actions', accepted for publishing by International Journal of Social Robotics. (Preprint available at arxiv.org/abs/2406.08824)
- Utilised interdisciplinary knowledge of ethical challenges in AI and human-robot interaction to engineer a robust set of prompts probing LLMs for bias in robotic decision-making settings. (Python)
- Promoted from Undergraduate Research Fellow to Research Assistant for commitment and excellent quality of work.

LEAD SOFTWARE ENGINEER **JAN 2024 – MAR 2024**

Student Records Management System Project for King's College London Informatics Department

- Developed a full-stack web application to detect discrepancies across multiple student records systems with hundreds of records each, automate mass email sending, and visualise student population metrics. (Python Django, JavaScript)
- Led client consultations, translating requirements into a system design and brand-aligned UI mockups using Figma.
- Acted as the 'backbone' of development, independently delivering features as well as integrating teammates' components to deploy a cohesive application under a tight deadline.
- Maintained a project-wide understanding to efficiently divide work between and provide technical guidance to teammates.
- Received thoroughly positive peer feedback, highlighting supportive leadership style, project-wide technical understanding, and meticulous code review practices.

Projects

OPEN-SOURCE LLM EVALUATION FRAMEWORK
Responsible Robotics & AI Lab
- Open-source framework to systematically evaluate LLMs for bias and discriminatory behaviors in robotic decision-making contexts, supports research paper accepted by International Journal of Social Robotics.
- Utilises efficient pipelines to probe LLMs with diverse prompt sets, generating results suitable for quantitative analysis.
- Published on GitHub to support research reproducibility: github.com/rumaisa-azeem/llm-robots-discrimination-safety
- Built with Python (HuggingFace Transformers, Pytorch).

COLLABORATIVE TASK MANAGEMENT APPLICATION
University Group Software Engineering Project
- Full-stack web application for team task management, allowing users to join multiple teams and track shared tasks.
- Features include user authentication system, notification system, searching and sorting of team tasks, assignment of status/due date/user to tasks.
- Built with Python (Django) and JavaScript.

PORTFOLIO WEBSITE
rumaisacodes.com
- Website to display professional portfolio and projects, with a focus on engaging and responsive frontend design.
- Built with React and TailwindCSS, deployed with Vercel.

LEADER OF WINNING HACKATHON TEAM
Global Power Hackathon 2022 with Pareto Economics, EY, and Microsoft
- Prototyped a working product dashboard and 5G-readiness index in a 12-hour time limit.
- Built with Python.

GUEST SPEAKER: PARALLELS BETWEEN AI AND NEUROSCIENCE
Philosophy & Science Society
- Delivered an hour-long lecture to an audience of 100+ members discussing the parallels between artificial intelligence and neuroscience after conducting extensive research into the topic.
- Speaker with highest audience attendance for the academic year – audience members reported the talk as inspiring, entertaining, and helpful for their understanding of the subject.

Skills

- **Languages**:
  - Python (4 years)
  - JavaScript & TypeScript (3 years)
  - C++ (2 years)
  - Java (familiar)
- **Full-stack frameworks:** Django, Flask, React (with Redux)
- **AI/ML:** PyTorch, TensorFlow, HuggingFace Transformers
- **Cloud SDKs:** AWS, GCP, Azure
- **Build & DevOps:** GitHub Actions, CMake, Docker
- **Testing:** Pytest, GoogleTest, Jest
- **Tools & practices:** Git, Linux, agile methodologies
- **UI & Design:** TailwindCSS, Figma